QuantMinds International

Stochastic Local Volatility vs Relative Entropy

two *perfect* models comparison

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Santander

Outline

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Introduction

This lecture addresses the comparison of Minimum Relative Entropy (MRE) and Stochastic Local Volatility (SLV) derivatives pricing methodologies.

In the world of financial derivatives, accurate pricing models are crucial for risk management, trading, and regulatory compliance.

Two advanced methodologies that have gained significant attention are the MRE and SLV models.

MRE Pricing is a sophisticated approach that seeks to find the pricing measure closest to the real-world probability measure by minimizing the relative entropy.

This method incorporates market information systematically and is particularly useful in incomplete markets where not all risks can be perfectly hedged.

On the other hand, the SLV Model combines the features of local volatility models and stochastic volatility models.



Introduction (ii)

SLV captures the dynamics of the underlying asset's volatility more accurately by allowing both local and stochastic components.

This hybrid model is especially effective in fitting market data and accurately model forward smile.

Throughout this lecture, we will delve into the theoretical foundations, mathematical formulations, and practical applications of both MRE and SLV models.

We will explore their advantages, disadvantages, and the contexts in which each model excels.

By the end of this session, you will have a comprehensive understanding of how these methodologies compare, enabling you to make informed decisions in the realm of derivatives pricing.

Let's begin our journey into the intricate world of MRE and SLV derivatives pricing methodologies.



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Measure Theory

Definition (Measurable space)

*Let Ω be a set and let $\mathcal{F}$ be a σ-algebra of its subsets, the pair $(\Omega, \mathcal{F})$ is said to be a **measurable space**.*

Definition (Measure, measure space)

*A **measure** is a mapping $\mu : \mathcal{F} \to [0, \infty]$ verifying the following properties:*

i. *Null empty set: $\mu(\varnothing) = 0$,*
ii. *Non-negativity: $\mu(A) \geq 0$ for all A in $\mathcal{F}$,*
iii. *Countable additivity or σ-additivity, for all countable collections $\{A_n\}_{n=1}^{\infty}$ of pairwise disjoint sets in $\mathcal{F}$:*

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu\left(A_n\right), \qquad \forall\, A_n \in \mathcal{F}.$$

*The triple $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**.*



Measure Theory and Probability

Definition (Absolutely continuous, equivalent measures)

Let μ and ν be two measures defined on $(\Omega, \mathcal{F})$, μ is said to be ***absolutely continuous*** *with respect to ν, denoted as $\mu \gg \nu$, if $\mu(A) = 0 \Rightarrow \nu(A) = 0$ for each $A \in \mathcal{F}$. If $\mu \gg \nu$ and $\nu \gg \mu$, μ and ν are said to be* ***equivalent measures****.*

Definition (Measurable function)

A mapping $X : \Omega \to \mathbb{R}$ is a ***measurable function*** *if $\forall\, B \in \mathcal{B}$ $X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$, where $\mathcal{B}$ is the Borel σ-algebra.*

Definition (Probability measure, probability space)

A measure $\mathbb{P}$ defined on a measurable space $(\Omega, \mathcal{F})$ verifying $\mathbb{P} : \mathcal{F} \to [0, 1]$, and $\mathbb{P}(\Omega) = 1$ is said to be a ***probability measure****, and $(\Omega, \mathcal{F}, \mathbb{P})$ is called a* ***probability space****.*

Definition (Random variable)

A measurable function $S : \Omega \to \mathbb{R}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is said to be a ***random variable****.*

Filtrations, Stochastic Processes and Martingales

Definition (Filtration)

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ a (discrete) ***filtration*** *is an increasing sequence of sub-σ-algebras, $\{\mathcal{F}_t\}_{t\geq 0}$ of $\mathcal{F}$, thus is*

$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \cdots \subset \mathcal{F}_t \subset \cdots \mathcal{F}.$$

The tuple $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ is called a **filtered probability space**.

Definition (Adapted, predictable, stochastic process)

A sequence of random variables $\{S_t = S_t(\omega)\}_{t\geq 0}$ is a ***stochastic process****. S_t is* ***adapted*** *if it is $\mathcal{F}_t$-measurable for all t, and it is said to be* ***predictable*** *if S_t is $\mathcal{F}_{t-1}$-measurable.*

Definition (Martingale)

An adapted stochastic process S_t is a ***martingale*** *regarding to $\mathcal{F}_t$, if it is integrable, and $\mathbb{E}^{\mathbb{P}}[S_T|\mathcal{F}_t] = S_t$ for each $t \leq T$ almost surely.*

When a property is fulfilled for each non zero measure set, it is said that it holds **almost surely (a.s.)**.



Portfolios and Trading Strategies

A **financial market** is composed by $M+1$ financial assets. $S_t^{(m)}$ denotes the m-th security price process for time $t \geq 0$.

Definition (Portfolio value, trading strategy)

*A **portfolio** is an adapted $(M+1)$-dimensional stochastic process, $\left(S_t^{(0)}, \ldots, S_t^{(M)}\right)$. The **portfolio value** process V_t is given by*

$$V_t := h_t^{(0)} S_t^{(0)} + \sum_{m=1}^{M} h_t^{(m)} S_t^{(m)}.$$

*Where each $h_t^{(m)}$ is a predictable stochastic process. The process $h_t^{(m)}$ expresses the number of units of $S_t^{(m)}$ held in the portfolio at time t and is said to be a **trading strategy**.*

Definition (Numeraire)

*A **numeraire** is any positive process used as reference for the portfolio value. The process $S_t^{(0)}$ is usually considered as the numeraire, and it is denoted by N_t.*



Self-financed Portfolio and Arbitrage Opportunity

Definition (Self-financed portfolio)

*A portfolio is said to be **self-financed** if its value process V_t verifies*

$$V_t - V_{t-1} = \sum_{m=0}^{M} h_t^{(m)} \left(S_t^{(m)} - S_{t-1}^{(m)} \right), \quad \text{for each} \quad t > 0.$$

Definition (Arbitrage opportunity)

*An **arbitrage opportunity** is a self-financed portfolio which verifies*

$$V_0 < 0, \quad V_T \geq 0, \tag{a}$$

or

$$V_0 = 0, \quad V_T \geq 0, \quad \mathbb{P}(V_T > 0) > 0. \tag{b}$$

Type (a) arbitrage is an immediate profit without future cost at T.

Whereas type (b) arbitrage requires no initial investment, has some possibility of a positive portfolio value at expiration T, and has no probability of terminal portfolio value being negative.



Derivative Securities and Complete Markets

The **basic market** comprises the aforementioned $M+1$ securities, however some additional securities are usually included too.

Definition (Derivative security payoff)

*A **derivative security payoff** on a filtered probability space $(\Omega, \mathcal{F}_t, \mathbb{P})$ is any random variable H measurable with respect to $\mathcal{F}_T$, i.e., it is generated by the basic market securities up to time T.*

The market including additional derivatives securities is said to be an **extended market**.

Definition (Replicable derivative security)

*A derivative security H is said to be **replicable** if there exists a self-financing portfolio V such that $H_T = V_T$.*

Definition (Complete market)

*The financial market is said to be **complete** if every derivative security can be replicated.*



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Fundamental Theorem and Martingale Pricing

Modern financial pricing is based on the absence of arbitrage notion, this is the goal of the following result.

Theorem 1 (Fundamental Theorem of Asset Pricing (FTAP))

A basic market on a filtered probability space $(\Omega, \mathcal{F}_t, \mathbb{P})$ *is* ***arbitrage-free****, if and only if* ***a martingale probability measure*** $\mathbb{Q}$ ***equivalent to*** $\mathbb{P}$ ***exists****, and it satisfies*

$$\frac{S_t^{(m)}}{N_t} = \mathbb{E}^{\mathbb{Q}}\left[\frac{S_T^{(m)}}{N_T}\middle|\mathcal{F}_t\right], \qquad \forall\, m \in \{1,\dots,M\},\ t \leq T. \tag{1}$$

Therefore there is no arbitrage opportunity if exists a probability measure $\mathbb{Q}$ for which the quotient between each security price process $S^{(m)}$ and the numeraire process N is a martingale.

Usual numeraire securities are the money-savings account, and the T-maturity zero coupon bond. Their associated martingale measures are called **risk-neutral measure**, and T-**forward measure** respectively.



Second Fundamental Theorem

The Fundamental Theorem of Asset Pricing (FTAP) tells nothing about market completeness or the uniqueness of the martingale measure, a result linking both concepts is given next.

Theorem 2 (Second FTAP)

*An arbitrage-free market is **complete** if and only if a **unique martingale measure** $\mathbb{Q}$ equivalent to $\mathbb{P}$ exists.*

The seminal reference addressing the Fundamental Theorem of Asset Pricing (FTAP) in discrete time for a finite securities market is Harrison and Kreps (1979).

Later Harrison and Pliska (1981; 1983), Taqqu and Willinger (1987), and Delbaen and Schachermayer (1994), extended and generalized both fundamental theorems.

The proofs of Theorems 1 and 2, for discrete time finite securities market, can be found in Capinski and E. Kopp (2012). For a general continuous time approach see Elliott and P. E. Kopp (2004).



Fundamental Theorems Interpretation

An intuitive understanding of previous theorems is that arbitrage-free derivatives valuation is an interpolation exercise.

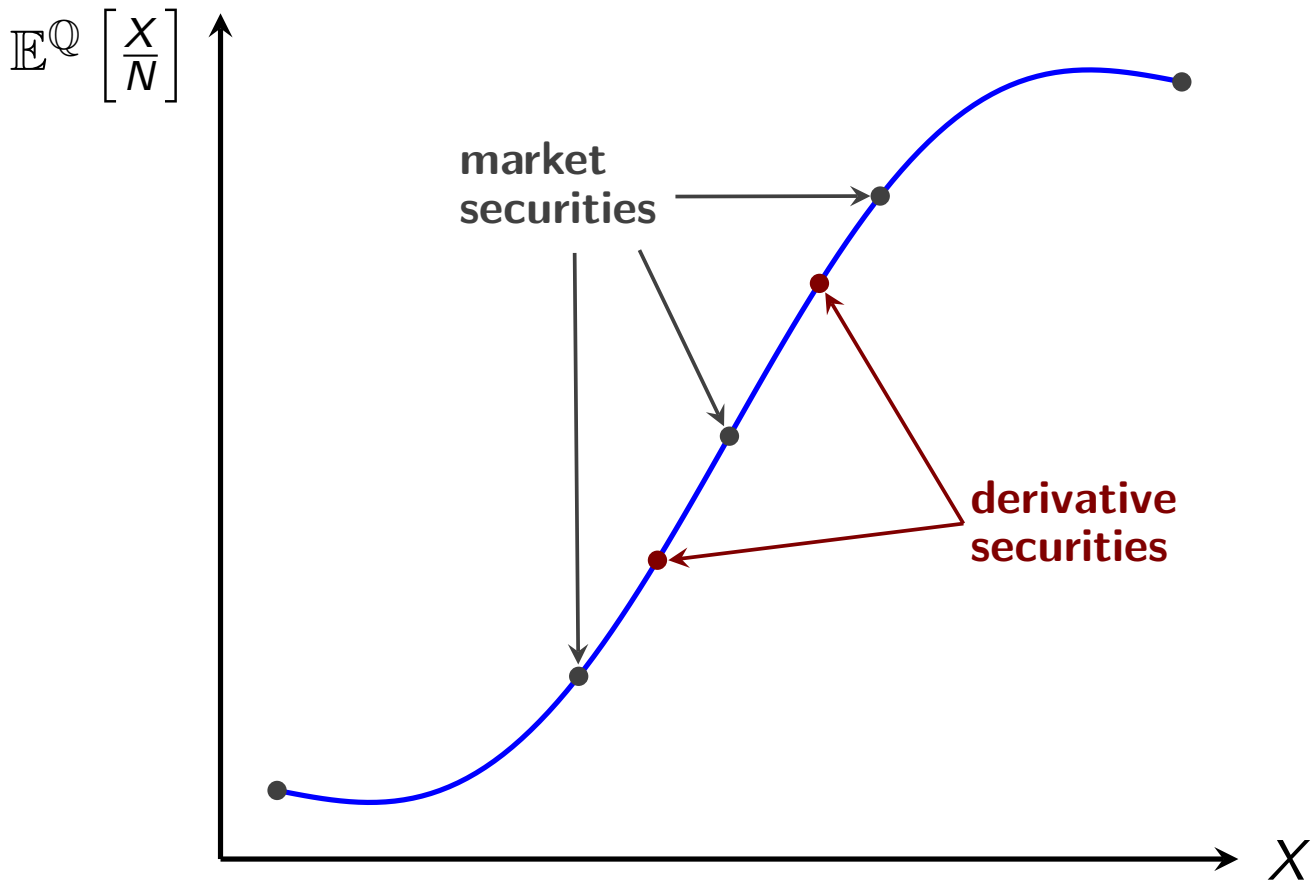


Figure 1: Martingale pricing in a complete market.

Martingale Measures and Incomplete Markets

Arbitrage-free pricing in an incomplete market can be regarded as a measure choice problem as illustrated below.

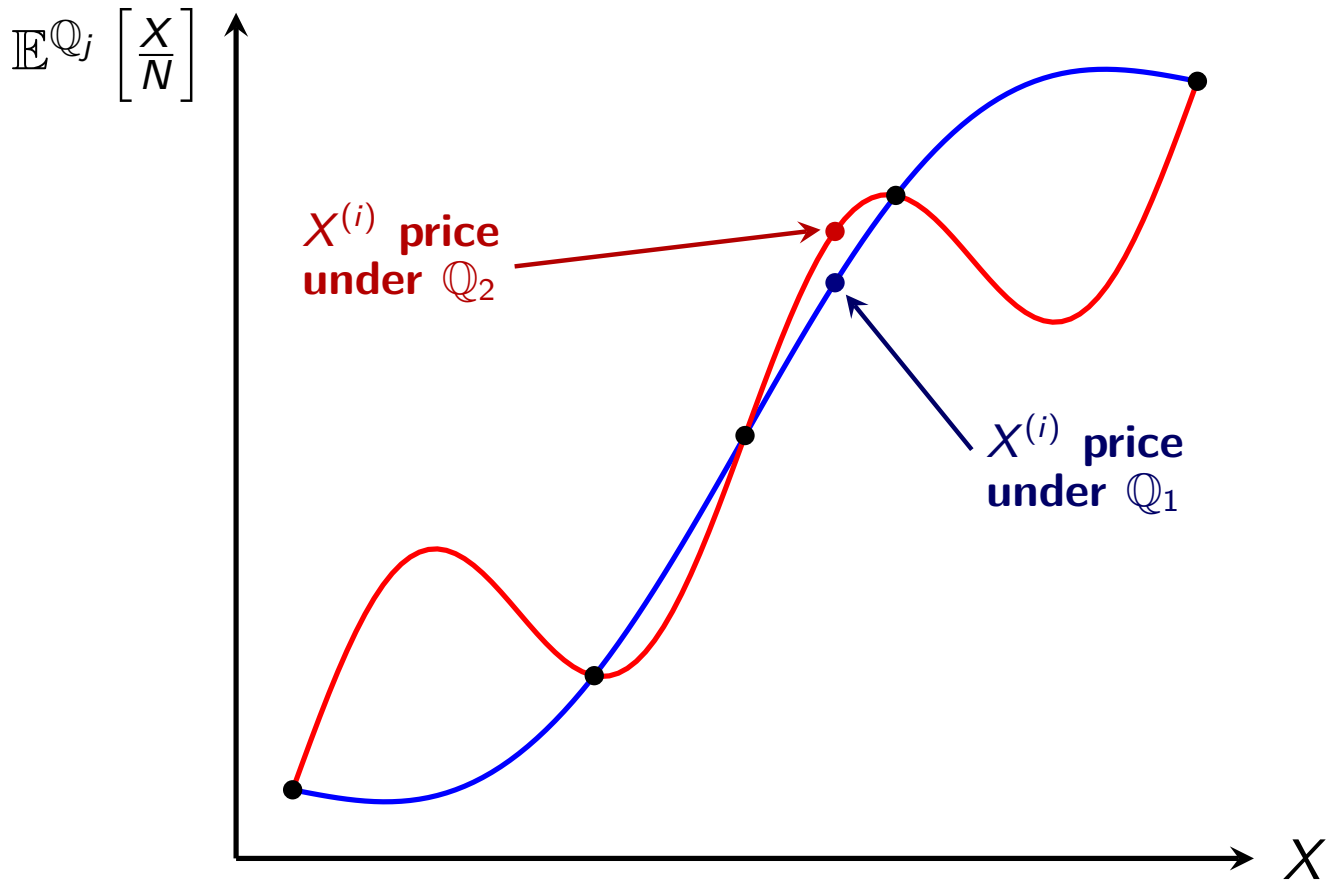


Figure 2: Martingale pricing in an incomplete market.

Feynman-Kac Theorem and Replication pricing

The replication approach for arbitrage-free pricing generalizes the well known Black and Scholes (1973) valuation framework.

Theorem 3 (Feynman-Kac)

Let x_t follow the stochastic differential equation (SDE)

$$\mathrm{d}x_t = a\left(x_t, t\right)\mathrm{d}t + b\left(x_t, t\right)\mathrm{d}W_t^{\mathbb{Q}},$$

where $W_t^{\mathbb{Q}}$ is Brownian motion under the measure $\mathbb{Q}$.

Let $V(x_t, t)$ be a differentiable function of x_t and t, which verifies the partial differential equation (PDE)

$$\frac{\partial V}{\partial t} + a\left(x_t, t\right)\frac{\partial V}{\partial x} + \frac{1}{2}b\left(x_t, t\right)^2\frac{\partial^2 V}{\partial x^2} - r\left(x_t, t\right)V(x_t, t) = 0,$$

with boundary condition $V(x_T, T)$, then the PDE has the solution

$$V(x_t, t) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(x_s, s)ds}V(x_T, T)\middle|\mathcal{F}_t\right]. \tag{2}$$

Feynman Kac Theorem and Martingale Pricing

A proof of Feynman-Kac theorem can be found for example in Capinski, E. Kopp, and Traple (2012).

The expectation in Feynman-Kac theorem is taken under the measure $\mathbb{Q}$, and its associated numeraire is $N_t = exp\left(-\int_0^T r\left(x_s, s\right) ds\right)$.

The Black and Scholes valuation framework is a particular case of the Feynman-Kac theorem where $a\left(x_t, t\right) = r\, x_t$, $b\left(x_t, t\right) = \sigma\, x_t$, and $V(x_T, T)$ is the payoff of an European vanilla option.

In this model, the $\mathbb{Q}$ is unique because the market is complete and the numeraire is a deterministic risk-free bank account.

The fundamental theorems and the Black-Scholes pricing framework are identical when the numeraire is the deterministic risk-free bank account and the market is complete.

In this setting, both valuation frameworks yield identical prices and can consequently be considered *"equivalent"* in their modeling assumptions.



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Radon-Nikodym Derivative

Suppose $\mathbb{P}$ and $\mathbb{Q}$ are equivalent measures. Given a path ω, for every ordered time mesh $\{t_1, \ldots, t_n\}$ (with $t_n = T$), we define x_i to be $W_{t_i}(\omega)$, and then the derivative $\frac{d\mathbb{Q}}{d\mathbb{P}}$ up to time T is defined to be the limit of the likelihood ratios

$$\frac{d\mathbb{Q}}{d\mathbb{P}}(\omega) = \lim_{n\to\infty} \frac{f^n_{\mathbb{Q}}(x_1, \ldots, x_n)}{f^n_{\mathbb{P}}(x_1, \ldots, x_n)},$$

as the mesh becomes dense in the interval $[0, T]$.

This continuous-time derivative $\frac{d\mathbb{Q}}{d\mathbb{P}}$ still satisfies the results that

$$\mathbb{E}^{\mathbb{Q}}[X_T] = \mathbb{E}^{\mathbb{P}}\left[\frac{d\mathbb{Q}}{d\mathbb{P}} X_T\right],$$

$$\mathbb{E}^{\mathbb{Q}}[X_t \mid \mathcal{F}_s] = \zeta_s^{-1}\mathbb{E}^{\mathbb{P}}[\zeta_t X_t \mid \mathcal{F}_s], \quad s \leqslant t \leqslant T,$$

where ζ_t is the process $\mathbb{E}^{\mathbb{P}}\left[\frac{d\mathbb{Q}}{d\mathbb{P}} \mid \mathcal{F}_t\right]$, and X_t is any process adapted to $\mathcal{F}_t$.



Radon-Nikodym Derivative (ii)

Just as the measure $\mathbb{P}$ can be approached through a limiting time mesh, so can the Radon-Nikodym derivative $\frac{d\mathbb{Q}}{d\mathbb{P}}$.

The event of paths agreeing with ω on the mesh,

$$A = \left\{\omega' : W_{t_i}\left(\omega'\right) = W_{t_i}(\omega), i = 1, \ldots, n\right\},$$

gets smaller and smaller till it is just the single point-set $\{\omega\}$.

The Radon-Nikodym derivative can be thought of as the limit

$$\frac{d\mathbb{Q}}{d\mathbb{P}}(\omega) = \lim_{A \to \{\omega\}} \frac{\mathbb{Q}(A)}{\mathbb{P}(A)}.$$

We have the mechanics of change of measure but still no clue about what change of measure does.

In fact all that measure changes on Brownian motion can do is to change the drift.

The subsequent result addresses this question.



The Cameron-Martin-Girsanov Theorem

Theorem 4 (Cameron-Martin-Girsanov (CMG))

If W_t is a $\mathbb{P}$-Brownian motion and γ_t is an $\mathcal{F}$-predictable process satisfying the boundedness condition

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2}\int_0^T \gamma_t^2 dt\right) < \infty\right],$$

then there exists a measure $\mathbb{Q}$ such that

(i) *$\mathbb{Q}$ is equivalent to $\mathbb{P}$;*

(ii) *$\frac{d\mathbb{Q}}{d\mathbb{P}} = \exp\left(-\int_0^T \gamma_t dW_t - \frac{1}{2}\int_0^T \gamma_t^2 dt\right)$; and*

(iii) *$\tilde{W}_t = W_t + \int_0^t \gamma_s ds$ is a $\mathbb{Q}$-Brownian motion.*

In other words, W_t is a drifting $\mathbb{Q}$-Brownian motion with drift $-\gamma_t$ at time t.

Within constraints, if we want to turn a $\mathbb{P}$-Brownian motion W_t into a Brownian motion with some specified drift γ_t, then there's a $\mathbb{Q}$ which does it, i.e., drift is measure and measure drift.



Cameron-Martin-Girsanov Converse

If W_t is a $\mathbb{P}$-Brownian motion, and $\mathbb{Q}$ is a measure equivalent to $\mathbb{P}$, then there exists an $\mathcal{F}$-predicatble process γ_t such that

$$\tilde{W}_t = W_t + \int_0^t \gamma_s ds$$

is a $\mathbb{Q}$-Brownian motion. That is, W_t plus drift γ_t is $\mathbb{Q}$-Brownian motion.

Additionally the Radon-Nikodym derivative of $\mathbb{Q}$ with respect to $\mathbb{P}$ (at time T) is

$$\exp\left(-\int_0^T \gamma_t dW_t - \frac{1}{2}\int_0^T \gamma_t^2 dt\right).$$

The CMG theorem applies to Brownian motion, but the majority of processes used in quantitative finance are "*disguised*" Brownian motions.



Cameron-Martin-Girsanov Converse (ii)

Suppose that X is a stochastic process with increment

$$dX_t = \sigma_t dW_t + \mu_t dt,$$

where W is a $\mathbb{P}$-Brownian motion.

Suppose we want to find if there is a measure $\mathbb{Q}$ such that the drift of process X under $\mathbb{Q}$ is $\nu_t dt$ instead of $\mu_t dt$.

As a first step, dX can be rewritten as

$$dX_t = \sigma_t \left(dW_t + \left(\frac{\mu_t - \nu_t}{\sigma_t} \right) dt \right) + \nu_t dt.$$

If we set

$$\gamma_t := (\mu_t - \nu_t) / \sigma_t,$$

and if γ then satisfies the CMG growth condition, i.e.,

$$\mathbb{E}^{\mathbb{P}} \left[\exp \left(\frac{1}{2} \int_0^T \gamma_t^2 dt \right) \right] < \infty.$$



Cameron-Martin-Girsanov Converse (iii)

Then indeed there is a new measure $\mathbb{Q}$ such that

$$\tilde{W}_t := W_t + \int_0^t \frac{\mu_s - \nu_s}{\sigma_s} ds,$$

is a $\mathbb{Q}$-Brownian motion.

But this means that the differential of X under $\mathbb{Q}$ is

$$dX_t = \sigma_t d\tilde{W}_t + \nu_t dt,$$

where $\tilde{W}$ is a $\mathbb{Q}$-Brownian motion - which gives X the drift ν_t we wanted.

Since the change of measure can only change the Brownian motion to a Brownian motion plus drift, the volatility of the process must remain the same.

For further detail see Baxter and Rennie (2012).



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Implied Volatility and the 1987 stock market crash

European options in the financial markets are sometimes quoted in terms of their so-called implied volatilities, instead of the commonly known bid and ask option prices.

The 1987 stock market crash, known as "*Black Monday*" occurred on October 19, 1987, when stock markets around the world plummeted.

The Dow Jones Industrial Average fell by 22.6% in a single day, marking one of the largest one-day percentage drop in history.

The crash had significant implications for the Black-Scholes model and option pricing in general.

The Black-Scholes model assumes constant volatility, which was challenged by the extreme market movements during the crash.

The sudden spike in implied volatility demonstrated that real market conditions can deviate significantly from model assumptions.

The crash led to the development of the volatility smile or skew, where implied volatility varies by strike price and expiration.



Implied Volatility Smile

This is roughly what implied volatilities looked like before and after the stock market crash of 1987.

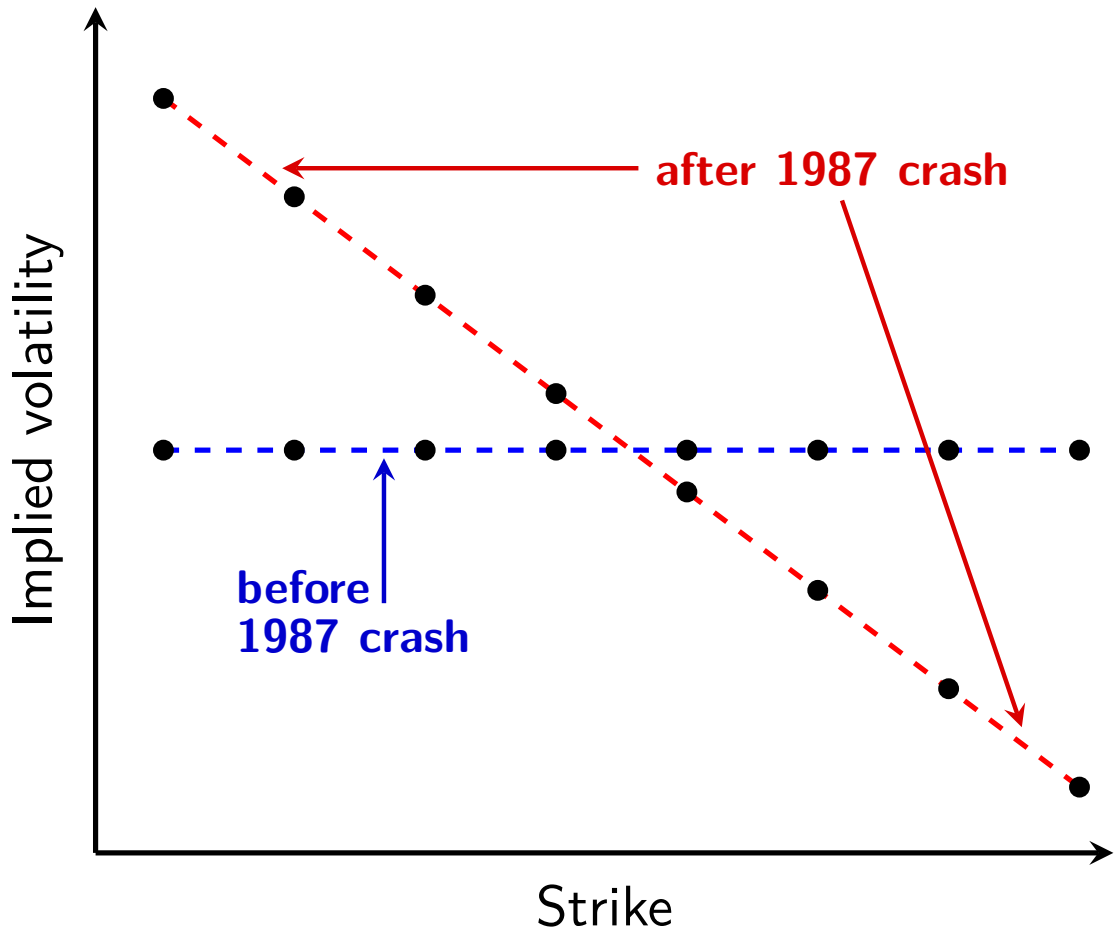


Figure 3: Quoted Black-Scholes volatility.

Implied Volatility Surface

The volatility surface is a three-dimensional graphical representation of implied volatility across different strike prices and expiration dates.

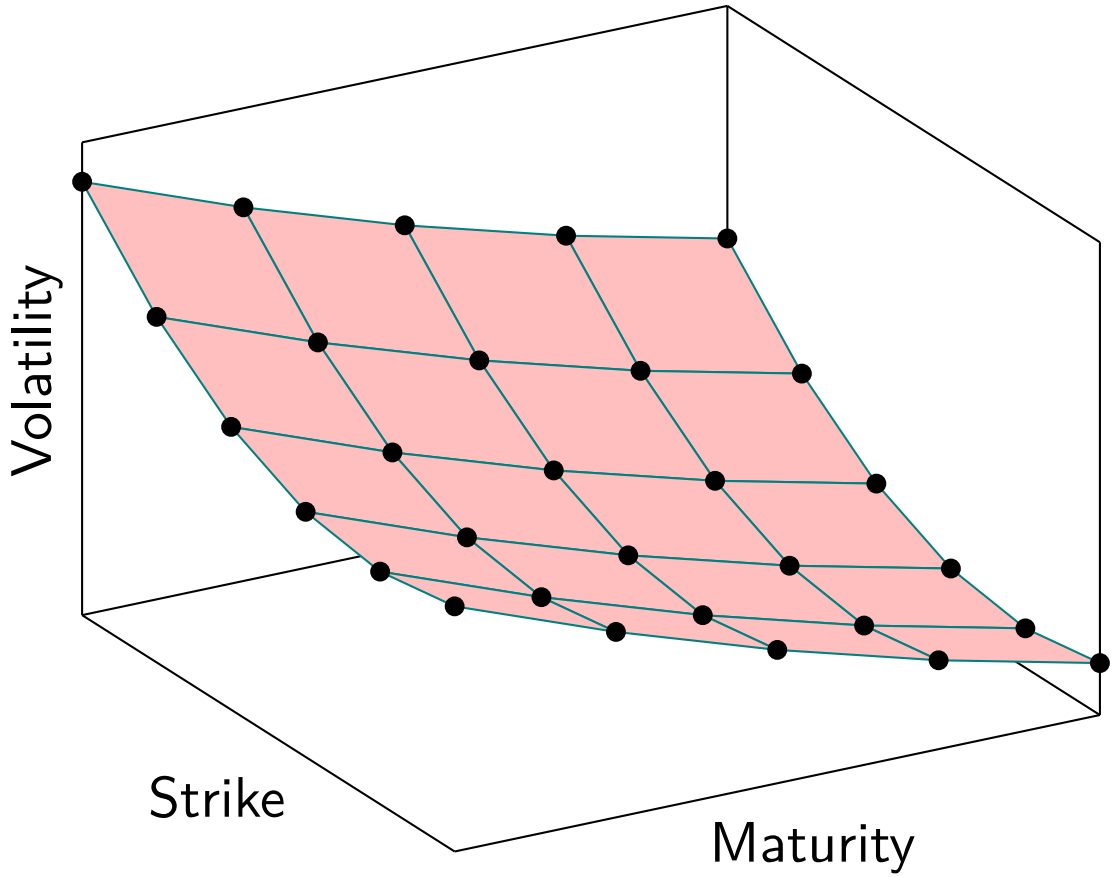


Figure 4: Implied volatility surface.

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Main Causes of the Volatility Surface

Non-Gaussian continuous returns. The assumption that Gaussian law serves as an effective model for the continuous return of the underlying asset is frequently contradicted by empirical data. Another common phenomenon observed is the so-called leverage effect, i.e., an asymmetry between positive and negative returns.

Continuous time hedging. A key requirement for achieving risk-free replication is the possibility of continuous-time hedging. If this is unattainable, there is a residual risk that the option seller will factor into the option's price, thereby impacting the volatility.

Transaction costs. Bid-ask spreads, different interest rates for funding or investment, etc. The presence of various transaction costs prevents risk-free hedging from being achieved. This translates into adjustments in the prices at which options are traded and consequently in their implied volatilities.

All of these issues stem from inherent limitations within the Black-Scholes modeling framework.



Other Drivers of the Volatility Surface

Supply and Demand. The balance between supply and demand for options at different strike prices and maturities affects the implied volatility, shaping the surface.

Market Sentiment: Investor expectations and market sentiment can cause fluctuations in implied volatility, leading to variations in the volatility surface.

Economic Events: Major economic events, such as earnings reports, interest rate changes, and geopolitical developments, can lead to sudden changes in implied volatility.

Historical Volatility: Past price movements and historical volatility patterns influence the market's expectations of future volatility, impacting the volatility surface.

Risk Preferences: The risk preferences of market participants, including their aversion to risk and their hedging strategies, play a role in determining the implied volatility.



Non-Gaussian Returns

The figure below presents the standardized daily log returns of the S&P 500 over the past four years, displayed together with its fitted Gaussian distribution.

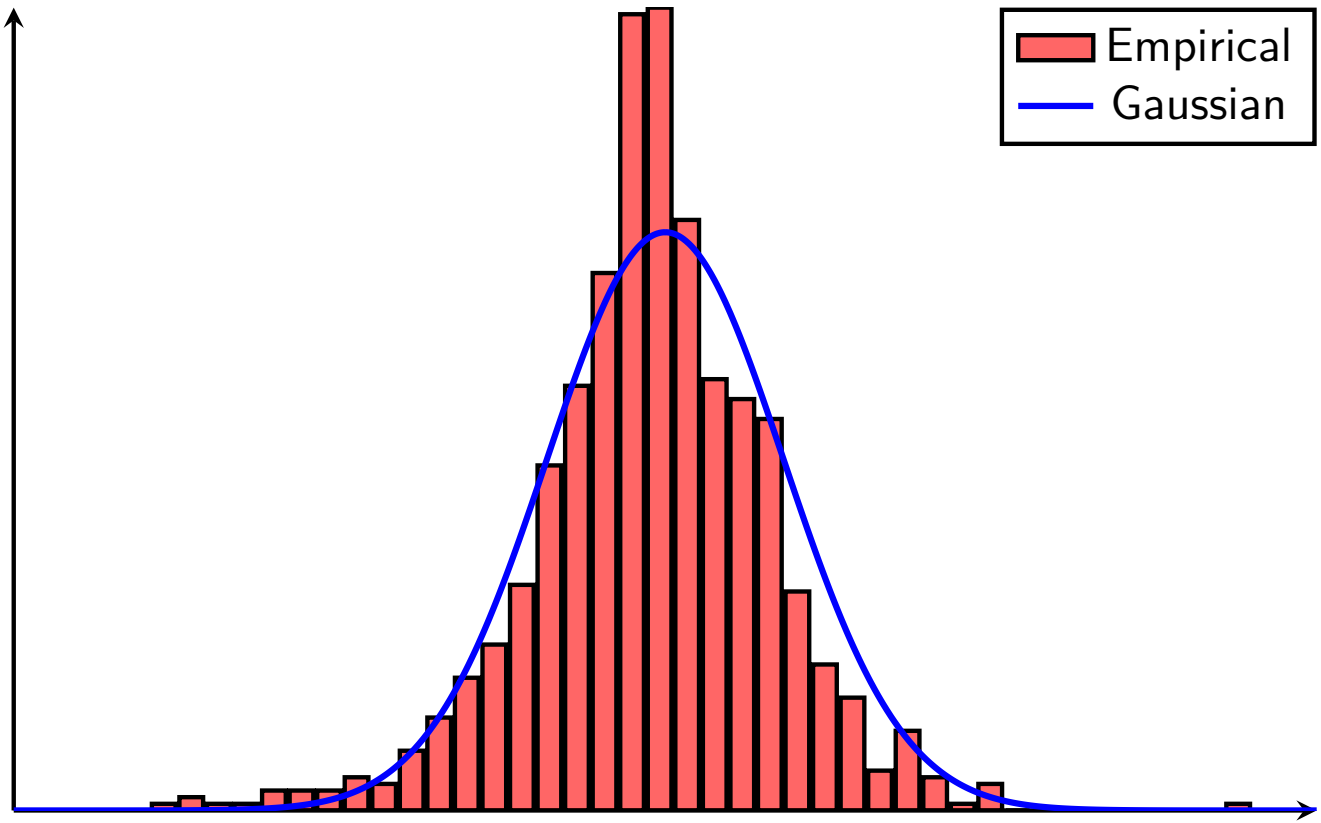


Figure 5: S&P 500 daily log returns and its Gaussian fit.

Empirical log returns left tail

In the following figure, tail events, i.e., returns that fall below the $\mu - 2\sigma$ threshold. This closer view highlights the frequency and impact of extreme downside movements.

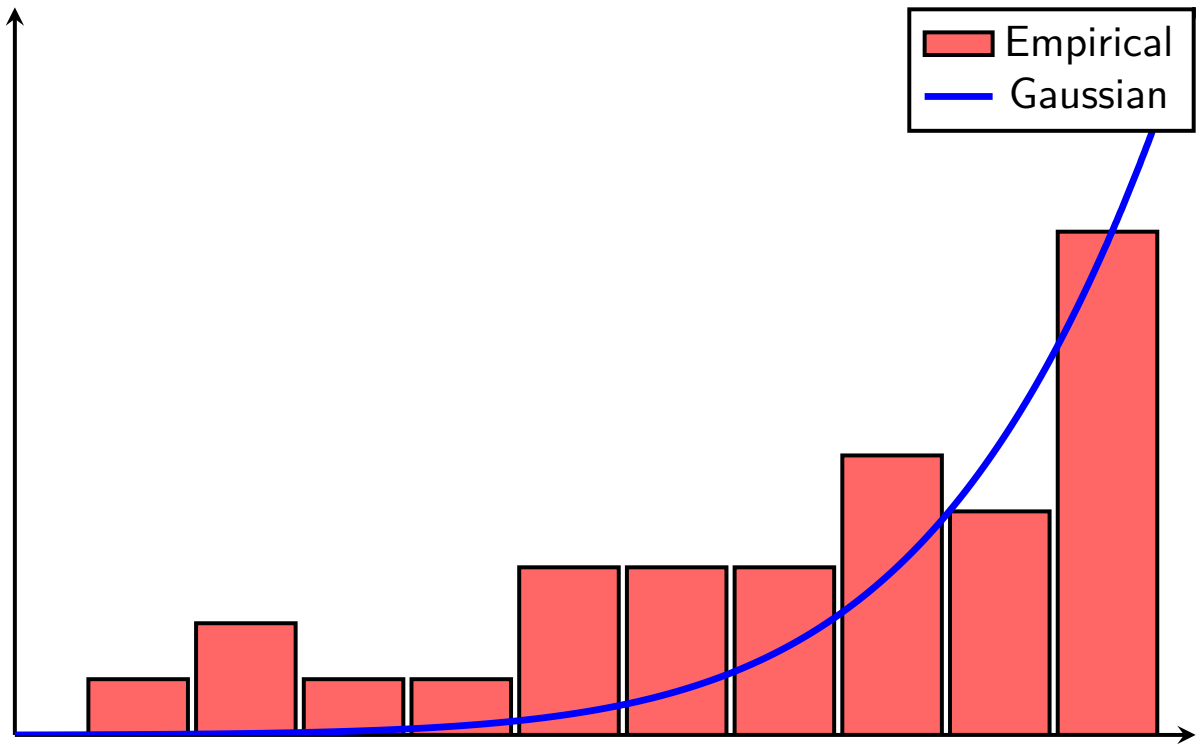


Figure 6: Left tail of the S&P 500 daily log returns distribution.

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Addressing the Volatility Surface Challenge

A straightforward extension to the Black-Scholes model, where σ is replaced by a deterministic function of time, $\sigma(t)$, partially addresses the issue.

With this adjustment, the model can price a single option for each maturity. However, it still lacks the flexibility to fully capture the entire volatility surface.

To address these limitations, practitioners often adjust the Black-Scholes model or use alternative models. Main modeling approaches are outlined below.

Jump-Diffusion Model pioneered by Merton (1976), incorporates both continuous and discontinuous price movements. This model employs a Wiener process to capture the continuous component of price movement and a Poisson process to model the occurrence of random jumps. These jumps, while not fully hedgeable individually, can be mitigated through diversification.



Addressing the Volatility Surface Challenge (ii)

Stochastic Volatility Models acknowledge that the volatility of financial assets is not constant but rather fluctuates over time. These models introduce a stochastic process to capture the time-varying nature of volatility, often influenced by the underlying asset's price movements. While Hull and White (1987) and E. M. Stein and J. C. Stein (1991) pioneered the concept, the Heston (1993) model has become an industry standard due to its analytical tractability.

Local Volatility Models, originally introduced by Dupire (1994), allow volatility to vary with the underlying asset price and time, matching the volatility surface. In the local volatility model, the volatility can be described by market quotes of option values, and it may be perfectly fit to the market option quotes while the model stays parameter-free.

Stochastic Volatility with Jumps. The Bates (1996) model combines a stochastic volatility with a jump component, allowing for both continuous price changes and sudden, large price shifts.



Addressing the Volatility Surface Challenge (iii)

Lévy Processes. Brownian motion, the Poisson, the compound Poisson processes and the jump diffusion process all belong to the finite activity Lévy processes, where paths typically consist of a continuous Brownian motion component and often a jump component. Examples of this approach for modeling the volatility surface include the Normal Inverse Gaussian model Barndorff-Nielsen (1997) or the Variance Gamma (VG) model Madan et al. (1998) among others.

Relative Entropy Minimization. As firstly proposed by Avellaneda (1998), the method involves an algorithm for calibrating asset-pricing models to the prices of benchmark securities. The algorithm computes the probability that minimizes the relative entropy with respect to a prior distribution and satisfies a finite number of moment constraints. These constraints arise from fitting the model to the prices of benchmark instruments, i.e., market prices. The algorithm is a special case of a general class of algorithms for calibrating models based on stochastic control and convex optimization.



Addressing the Volatility Surface Challenge (iv)

Stochastic Local Volatility Models. It is a combined model aiming to take advantage of the benefits of the individual models, and remove as much as possible their limitations, the class of stochastic-local volatility models was developed by Jex et al. (1999), Lipton (2002) and Lipton and McGhee (2002), amongst others.

Parametric Volatility Surface calibrated directly from market data, constructing a bespoke volatility surface for pricing and hedging. Main examples are Hagan et al. (2002) Stochastic Alpha Beta Rho (SABR), and Gatheral (2004) Stochastic Volatility Inspired (SVI) models.

Fractional Brownian Motion and Rough Volatility models are also a generalization of Brownian motion allowing increments to be not independent of each other. Fractality means that the shape of an object doesn't change its form at different time scales. This is a very important feature in analyzing volatility behavior. This modeling approach for the volatility was initially proposed by Gatheral et al. (2018) and named Rough Volatility.



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Heston Stochastic Volatility Model

In stochastic volatility models two stochastic differential equations, one for the underlying asset price $S(t)$, and one for the variance process $v(t)$ are used.

In the Heston (1993) stochastic volatility model, these two stochastic differential equations, under the risk-neutral measure $\mathbb{Q}$, are

$$\begin{cases} \mathrm{d}S(t) = rS(t)\mathrm{d}t + \sqrt{v(t)}S(t)\mathrm{d}W_s^{\mathbb{Q}}(t), \\ \mathrm{d}v(t) = \kappa\Big(\theta - v(t)\Big)\mathrm{d}t + \gamma\sqrt{v(t)}\mathrm{d}W_v^{\mathbb{Q}}(t), \end{cases} \tag{3}$$

with initial conditions $S\left(t_0\right) = S_0 > 0$, and $v\left(t_0\right) = v_0 > 0$, respectively. Additionally, a correlation is defined between the underlying Brownian motions, i.e.,

$$\mathrm{d}W_s^{\mathbb{Q}}(t)\mathrm{d}W_v^{\mathbb{Q}}(t) = \rho_{s,v}\,\mathrm{d}t.$$

Parameters $\kappa \geq 0$, $\theta \geq 0$, and $\gamma > 0$ are called the speed of mean reversion, the long-term mean of the variance process and the volatility of the volatility, respectively.



Variance Process in the Heston Model

The variance process in the Heston model is a mean-reverting Cox-Ingersoll-Ross (CIR) process originally proposed by Cox et al. (1985) to model interest rates.

The variance CIR process is defined, under $\mathbb{Q}$-measure, by the second SDE of (3). It is a mean reverting square-root process which precludes negative values for $v(t)$, and if $v(t)$ reaches zero it subsequently becomes positive.

The Feller condition, i.e., $2\kappa\theta \geq \gamma^2$, guarantees that $v(t)$ stays positive, if this condition is not verified, $v(t)$ may reach zero.

When the Feller condition is not fulfilled, the cumulative distribution exhibits a near-singular behavior near the origin.

If so, there is an accumulation of probability mass around zero and the density needs to become extreme in order to fit the tail near the singularity around zero with the continuous part of the density.

See Feller (1951) for further details.



$\mathbb{P}$ Measure and the Heston Model

Heston dynamics under the real-world $\mathbb{P}$-measure is is formulated as:

$$\begin{cases} \mathrm{d}S(t) = \mu S(t)\mathrm{d}t + \sqrt{v(t)}S(t)\mathrm{d}W_s^{\mathbb{P}}(t), \\ \mathrm{d}v(t) = \kappa\left(\theta^{\mathbb{P}} - v(t)\right)\mathrm{d}t + \gamma\sqrt{v(t)}\mathrm{d}W_v^{\mathbb{P}}(t), \end{cases}$$

with the correlation between the Brownian motions under the $\mathbb{P}$-measure given by $\mathrm{d}W_s^{\mathbb{P}}(t)\mathrm{d}W_v^{\mathbb{P}}(t) = \rho_{s,v}\ \mathrm{d}t$.

The following measure transformation, Oosterlee and Grzelak (2020, Chapter 8), can be found:

$$\begin{aligned} \mathrm{d}\widetilde{W}_x^{\mathbb{P}}(t) &= \mathrm{d}\widetilde{W}_x^{\mathbb{Q}}(t) - (\mu - r)/\sqrt{v(t)}\mathrm{d}t, \\ \mathrm{d}\widetilde{W}_v^{\mathbb{P}}(t) &= \mathrm{d}\widetilde{W}_v^{\mathbb{Q}}(t). \end{aligned}$$

With this measure transformation, it is necessary to satisfy $v(t) > 0$.

By setting $\theta := \theta^{\mathbb{P}} - \rho_{s,v}\gamma(\mu - r)/\kappa$, the system of SDE's (3) in the $\mathbb{Q}$-measure is obtained.



Martingale Approach in the Heston model

For the Heston model (3), the option pricing problem is formulated as:

$$\mathbb{E}^{\mathbb{Q}}\left[\frac{V(S,v,T)}{N(T)}\middle|\mathcal{F}_t\right]=\frac{V(S,v,t)}{N(t)}.$$

Assume a differentiable function, the discounted option value, which should be a martingale. Its dynamics using Itô's lemma can be determined, i.e.,

$$\mathrm{d}\left(\frac{V}{N}\right)=\frac{1}{N}\,\mathrm{d}V-r\frac{V}{N}\,\mathrm{d}t,$$

and for an infinitesimal increment, $\mathrm{d}V$,

$$\mathrm{d}V=\left(\frac{\partial V}{\partial t}+rS\frac{\partial V}{\partial S}+\kappa(\theta-v)\frac{\partial V}{\partial v}+\frac{1}{2}vS^2\frac{\partial^2 V}{\partial S^2}+\rho_{s,v}\gamma Sv\frac{\partial^2 V}{\partial S\partial v}\right.$$
$$\left.+\frac{1}{2}\gamma^2 v\frac{\partial^2 V}{\partial v^2}\right)\mathrm{d}t+S\sqrt{v}\frac{\partial V}{\partial S}\,\mathrm{d}W_x+\gamma\sqrt{v}\frac{\partial V}{\partial v}\,\mathrm{d}W_v.$$

Martingale Approach in the Heston model (ii)

For the martingale property to hold, the dynamics should be free of $\mathrm{d}t$-terms, which yields

$$\begin{aligned}\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \kappa(\theta - v)\frac{\partial V}{\partial v} + \frac{1}{2}vS^2\frac{\partial^2 V}{\partial S^2} \\ + \rho_{s,v}\gamma Sv\frac{\partial^2 V}{\partial S \partial v} + \frac{1}{2}\gamma^2 v\frac{\partial^2 V}{\partial v^2} - rV = 0,\end{aligned} \tag{4}$$

i.e., the Heston PDE, for derivatives pricing.

Solutions to (4) can be found with the help of the Feynman-Kac theorem, by which we can describe the solution as the discounted expected payoff under the risk-neutral measure.

The conditional density function in the Feynman-Kac representation is not known, but its discounted characteristic function can be determined, yielding a semi-analytical solution for the derivative price.

This property facilitates the calibration of the model to market data.



Replication and Hedging with Stochastic Volatility

The market completeness rule of thumb is that the number of assets and the number of sources of randomness (i.e., Brownian motions) should be the same.

The situation is nontrivial when considering stochastic volatility processes.

The difficulty is related to the fact that the volatility itself cannot easily be bought or sold on the market, this is, it is not a unique measurable and traded quantity.

In order to hedge both the risk associated with S and the random variance v, two hedging instruments are needed.

Consequently, two hedging ratios need to be defined. The first one is Δ_s units of the underlying asset, and the second one is Δ_v units of another derivative, which is bought with value U.

This way we may deal with the two sources of randomness, i.e., in $S(t)$ and in $v(t)$.



Replication and Hedging with Stochastic Volatility (ii)

As a consequence, the hedging portfolio Π of the target derivative V, is given by:

$$\Pi(S,t,v) = V(S,t,v) - \Delta_s S(t) - \Delta_v U(S,t,v),$$

where $U(S,t,v)$ is the value of the hedging instrument, and the dependence on S, t, and v has being explicitly included in the arguments of Π, V, and U.

It can be shown that by prescribing:

$$\frac{\partial V}{\partial S} - \Delta_s - \Delta_v \frac{\partial U}{\partial S} = 0,$$

and

$$\frac{\partial V}{\partial v} - \Delta_v \frac{\partial U}{\partial v} = 0,$$

the randomness from the portfolio dynamics is eliminated.



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Local Volatility Model

The classical local volatility model, firstly presented by Dupire (1994), is given by

$$\mathrm{d}S(t) = rS(t)\mathrm{d}t + \sigma_{\mathrm{LV}}(S,t)S(t)\mathrm{d}W(t),$$

with initial condition $S(t_0) = S_0$, and a constant or deterministic interest rate r.

For a given option value $V(S,t)$, the no-arbitrage assumption gives rise to the following partial differential equation (PDE)

$$\begin{cases} \dfrac{\partial V}{\partial t} + \dfrac{1}{2}\sigma_{\mathrm{LV}}^2(S,t)S^2\dfrac{\partial^2 V}{\partial S^2} + rS\dfrac{\partial V}{\partial S} - rV = 0, \\ V(S,T) = \max\Big(S(T) - K, 0\Big). \end{cases} \tag{5}$$

The PDE (5) is a backward Kolmogorov equation that describes how the option value $V(S,t)$ evolves over time.

The subsequent results will be essential for calculating σ_{LV}^2.



Market Prices and Risk-Neutral Density

The stock density is directly linked to the market quotes of options through the well-known result established by Breeden and Litzenberger (1978).

The Breeden-Litzenberger formula is

$$f_{S_T}(K) = e^{-r(T-t_0)} \frac{\partial^2 V_c(K,T)}{\partial K^2}, \tag{6}$$

where $f_{S_T}(K)$ denotes the risk-neutral density of the stock price process at time T, and $V_c(K,T)$ represents the market call option prices at time t_0, with an initial stock value of S_0, a strike price of K, and a maturity of T.

It is worth noticing that only a limited number of options are quoted in the market for a particular stock.

Consequently, accurately recovering the stock density from these limited market quotes is challenging, as an infinite number of options would be required across the strike dimension $K \in [0, \infty)$.



General SDE and Kolmogorov PDE

A general SDE's probability density function (PDF) can be derived from the Kolmogorov Forward PDE, which governs its temporal evolution.

Theorem 5 (General SDE and Kolmogorov Forward PDE)

Let x_t follow the stochastic differential equation (SDE)

$$\mathrm{d}x(t) = a(x,t)\mathrm{d}t + b(x,t)\mathrm{d}W(t), \quad t_0 \leq t \leq T,$$

with initial condition $x(t_0) = x_0$, and where W_t is a Brownian motion. Then the transition density $f(x,t)$ associated to the above SDE satisfies the Kolmogorov Forward PDE:

$$\begin{cases} \dfrac{\partial}{\partial t} f(x,t) + \dfrac{\partial}{\partial x}\left[a(x,t) f(x,t)\right] - \dfrac{1}{2}\dfrac{\partial^2}{\partial x^2}\left[b^2(x,t) f(x,t)\right] = 0, \\ f(x,t_0) = \delta\left(x = x_0\right). \end{cases} \tag{7}$$

This PDE is also known as the Fokker-Planck forward equation. The proof of this result can be found in Karatzas and Shreve (1991).



Local Variance Function

By setting $a(x,t) = S \cdot \sigma_{\text{LV}}(S,t)$, in the SDE from Theorem 5, and utilizing Breeden-Litzenberger formula (6), the expression for the local volatility function $\sigma_{\text{LV}}(K,T)$ is derived as follows

$$\sigma^2_{\text{LV}}(K,T) = \frac{\dfrac{\partial V_c(K,T)}{\partial T} + rK\dfrac{\partial V_c(K,T)}{\partial K}}{\dfrac{1}{2}K^2\dfrac{\partial^2 V_c(K,T)}{\partial K^2}}, \tag{8}$$

where again $V_c(K,T)$ is the market call option price at time t_0, initial stock value S_0, strike price K and maturity T.

The local volatility $\sigma_{\text{LV}}(K,T)$ can be characterized by the market quotes of option values. This allows the model to perfectly fit the market option quotes while remaining parameter-free.

In other words, there is no requirement for a calibration procedure, simplifying the modeling process and enhancing its applicability.



Local Volatility in Practice

From Equation (8), it is evident that the local volatility $\sigma_{\text{LV}}(K, T)$ can be derived from European option prices.

To compute this so-called Dupire's local volatility term, an option price surface is required. Using the local volatility model, we can then price options, for instance, through the finite difference method applied to the resulting option valuation PDE.

In practice, however, not all partial derivatives in (8) can be directly obtained from market option quotes; some may require calculation via finite difference approximations using market data.

For each local volatility term, four option values are necessary for these finite difference approximations of the partial derivatives.

Additionally, in the denominator of (8), we encounter an approximation of the stock price density.

Here, finite difference approximations can introduce significant errors, particularly for high strike prices.



Local Volatility in Practice (ii)

The errors can become particularly pronounced when the stock density approaches zero, leading to unrealistically high values for the local volatility function $\sigma_{\text{LV}}(K, T)$.

To enhance the accuracy of local volatility, one effective approach is to parameterize the option surface using a two-dimensional polynomial function $g(K, T)$ and fit it to market data.

Notable examples of this approach include the SABR model by Hagan et al. (2002) and the SVI model by Gatheral (2004).

In this method, the partial derivatives of $g(K, T)$ with respect to time and strike price can be obtained analytically. However, a significant challenge arises in selecting a parametric function that accurately fits all market quotes while adhering to no-arbitrage principles.

Moreover, fitting a polynomial to option prices can pose difficulties, especially for out-of-the-money (OTM) options. Small variations in prices can lead to substantial discrepancies in implied volatilities.



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Hybrid Modeling Approach

The Heston model is renowned for generating an implied volatility forward smile that closely mirrors the observed implied volatility smile in today's financial markets. This alignment facilitates accurate pricing of products sensitive to forward implied volatility.

However, calibrating the Heston model can be quite challenging, and it is important to note that not all types of implied volatility smiles and skews can be effectively captured by its dynamics.

The Local Volatility model offers a precise fit to current market option quotes, ensuring that the model aligns perfectly with observed prices. Nevertheless, it tends to flatten the forward implied volatility smile.

To leverage the strengths of both models while mitigating their weaknesses, the class of stochastic-local volatility models were introduced by Jex et al. (1999), Lipton (2002) and Lipton and McGhee (2002).

The subsequent slides will follow the methodology outlined in Oosterlee and Grzelak (2020, Chapter 10).



Stochastic Local Volatility Model

The stochastic-local volatility (SLV) model under measure $\mathbb{Q}$ is governed by the following system of SDEs:

$$\begin{cases} \dfrac{\mathrm{d}S(t)}{S(t)} = r\ \mathrm{d}t + \eta(S,t)\xi\left(v(t)\right)\mathrm{d}W_s(t), \\ \mathrm{d}v(t) = a_v(v,t)\mathrm{d}t + b_v(v,t)\mathrm{d}W_v(t), \\ \mathrm{d}W_s(t)\mathrm{d}W_v(t) = \rho_{s,v}\ \mathrm{d}t, \end{cases} \tag{9}$$

where $\rho_{s,v}$ denotes the correlation between the Brownian motions, and r is the constant interest rate.

The function $\eta(S,t)$ is the local volatility component, while $\xi(v)$ controls the stochastic volatility, the terms $a_v(v,t)$ and $b_v(v,t)$ represent the drift and diffusion of the variance process, respectively.

Two stochastic volatility models that fit into this framework are the previously discussed Heston (1993) model and the Schöbel and Zhu (1999) model.



Local Volatility Component

In the SLV framework there is thus one free function available, i.e., $\eta(S,t)$ and this local volatility component may be defined such that the densities implied from the market and the model are equal.

The following relation, see Oosterlee and Grzelak (2020, Section 10.2), is obtained

$$\eta(K,t) = \frac{\sigma^2_{\text{LV}}(K,t)}{\mathbb{E}^{\mathbb{Q}}\left[\xi^2(v(t))\middle|\, S(t)=K\right]}, \tag{10}$$

where the expectation is conditioned on $\mathcal{F}_{t_0}$.

The SLV local volatility component $\eta(K,t)$ thus consists of two components, a deterministic local volatility $\sigma_{\text{LV}}(K,t)$ and a conditional expectation $\mathbb{E}^{\mathbb{Q}}\left[\xi^2(v(t))\middle|\, S(t)=K\right]$.

The efficient computation of the conditional expectation presents a significant challenge. The complexity arises from the fact that the joint distribution of the variance v and the stock S is unknown and cannot be directly inferred from market quotes.



Heston SLV Model Monte Carlo Simulation

Under $\mathbb{Q}$ the Heston SLV model evolves according to

$$\begin{cases} \dfrac{\mathrm{d}S(t)}{S(t)} = r\mathrm{d}t + \eta(S,t)\sqrt{v(t)}\mathrm{d}W_s(t), \\ \mathrm{d}v(t) = \kappa\Big(\theta - v(t)\Big)\mathrm{d}t + \gamma\sqrt{v(t)}\mathrm{d}W_v(t), \end{cases}$$

with $\rho_{s,v}$ the correlation between the $\mathrm{d}W_s(t)$ and $\mathrm{d}W_v(t)$ processes.

An Euler discretization of the Heston SLV model is as follows

$$s_{i,j+1} = s_{i,j} + r\, s_{i,j}\Delta t + \sqrt{\frac{\sigma_{\mathrm{LV}}^2\left(t_i, s_{i,j}\right)}{\mathbb{E}^{\mathbb{Q}}\left[v(t_j)\middle|\, S(t_j) = s_{i,j}\right]}}\, s_{i,j}\sqrt{v_{i,j}\Delta t}\, X,$$

$$v_{i,j+1} = v_{i,j} + \kappa\left(\theta - v_{i,j}\right)\Delta t + \gamma\sqrt{v_{i,j}\Delta t}\,\left(\rho_{s,v}X + \sqrt{1-\rho_{s,v}^2}\,Y\right),$$

where i indexes the paths (1 to n), j indexes the time steps (0 to m), X and Y are independent standard Gaussian random variables, and $\Delta t := \frac{T}{m}$ is the time step size with T being the time horizon.



Conditional Expectation Evaluation

To compute the asset path values at the subsequent time step, t_{j+1}, precise approximation of the two components, $\sigma^2_{\mathrm{LV}}(t_i, s_{i,j})$ and $\mathbb{E}^{\mathbb{Q}}[v(t_j)|\, S(t_j) = s_{i,j}]$, is essential.

Efficient evaluation of $\sigma^2_{\mathrm{LV}}(t_i, s_{i,j})$ can be achieved through various methods, including parametric forms and finite differences.

Directly extracting $\mathbb{E}^{\mathbb{Q}}[v(t_j)|\, S(t_j) = s_{i,j}]$ from market quotes is not feasible. Estimating this term is challenging due to the need for conditional evaluation for each individual stock realization, $s_{i,j}$.

In the discretization of (S, v) each realization of $s_{i,j}$ corresponds uniquely to a specific realization of the variance $v_{i,j}$. This one-to-one relationship complicates the evaluation of the conditional expectation, making it a more intricate task.

To tackle the challenging conditional expectation, Stoep et al. (2014) proposed a nonparametric Monte Carlo method which relies on splitting the Monte Carlo realizations in bins.



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Random Variable Uncertainty

Entropy is directly related to the uncertainty of the outcome of a random variable.

Definition (Entropy of a random variable)

Let X be a discrete random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, then

$$H(X) := -\mathbb{E}^{\mathbb{P}}[\log p(\omega)] = -\sum_{\omega \in \Omega} p(\omega) \log p(\omega)$$

*is said to be the **entropy** of the random variable X.*

Entropy main properties are:

i. if there is no uncertainty then the entropy is zero;

ii. entropy only depends on the probability measure; and

iii. largest entropy value if $p(\omega)$ is equal for all $\omega \in \Omega$.

For example, if X is a discrete uniform random variable, its entropy is calculated as $\log n$, where n is the number of possible outcomes.



Relative Entropy

The concept of entropy can be extended as the next definition states.

Definition (Relative Entropy)

Let X, Y be two discrete random variable on probability spaces $(\Omega, \mathcal{F}, \mathbb{P})$, and $(\Omega, \mathcal{F}, \mathbb{Q})$ respectively, then

$$D\left(\mathbb{Q}||\mathbb{P}\right) := \mathbb{E}^{\mathbb{Q}}\left[\log \frac{q(\omega)}{p(\omega)}\right] = \sum_{\omega \in \Omega} q(\omega) \log \frac{q(\omega)}{p(\omega)},$$

*is said to be the **relative entropy** of the two random variables.*

It quantifies the difference of starting with $\mathbb{P}$ probability measure and ending with $\mathbb{Q}$, its main properties are:

i. it is non-negative;

ii. $D\left(\mathbb{Q}||\mathbb{P}\right) = 0$ iff $\mathbb{P} = \mathbb{Q}$; and

iii. it is not symmetric and neither satisfies the triangle inequality.

It is also referred as Kullback-Leibler divergence.



Minimum Relative Entropy

Minimum Relative Entropy (MRE) was initially proposed for asset pricing by Avellaneda (1998).

The idea is using relative entropy as the objective function in the following constrained optimization problem.

$$\min_{\mathbb{Q}\in\mathcal{Q}} D\left(\mathbb{Q}||\mathbb{P}\right)$$

subject to: (11)

$$\mathbb{E}^{\mathbb{Q}}\left[g_j(\omega)\right] = c_j, \qquad j = 1, 2, \ldots, m,$$

where g_j and c_j are given random variables and constants respectively.

In our current context g_j is the j-th security value divided by the numeraire, and c_j is its market price also divided by the numeraire.

If the solution to the problem (11) exists, then it can be found by the Lagrange multipliers technique.

Technical details can be found in Cover and Thomas (2006).



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Monte Carlo and Minimum Relative Entropy

Avellaneda et al. (2001) proposed the use of a Monte Carlo simulation as starting, also called prior, probability measure $\mathbb{P}$.

Each Monte Carlo path is an event ω and there are n of these paths, therefore $p(\omega) = \frac{1}{n}$ for each $\omega \in \Omega$.

This allows rewriting the constrained optimization problem (11) as

$$\min_{\mathbb{Q} \in \mathcal{Q}} \sum_{i=1}^{n} q_i \log q_i$$

subject to: (12)

$$\sum_{i=1}^{n} g_{ij} q_i = c_j, \qquad j = 1, 2, \ldots, m,$$

where q_i is path i probability under $\mathbb{Q}$, and g_{ij} denotes the path i ratio between the j-th security and the numeraire.

Notice that $\log n$ has been suppressed from the objective function, since it is a constant as $p_i := p(\omega) = 1/n$ for each $\omega \in \Omega$.



Weighted Monte Carlo

The solution of the latter constrained optimization problem is obtained by Lagrange multipliers, yielding the solution

$$q_i = \frac{\exp\left(\sum_{j=1}^{m} \lambda_j^* g_{ij}\right)}{\sum_{i=1}^{n} \exp\left(\sum_{j=1}^{n} \lambda_j^* g_{ij}\right)}, \tag{13}$$

where λ_j^* is the optimal value of the j-th Lagrange multiplier solving the constrained optimization problem (12).

A very useful algorithm for a numerical resolution of the optimization problem is given in Elices and Gimenez (2006).

Standard Monte Carlo simulation assigns the weight $\frac{1}{n}$ to each of the n simulated paths.

The MRE posterior measure given by the q_i's can be considered as a new *weight* assigned to each path, hence the name of Weighted Monte Carlo (WMC) firstly coined by Avellaneda et al. (2001).



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MRE Pricing Framework

A very general pricing framework can be stablished by means of the MRE technique as presented in Arrieta (2015).

It must take as inputs: a model, i.e., a prior measure $\mathbb{P}$; and a set of benchmark securities; It should output a new measure $\mathbb{Q}$, that is, a *new pricing model*.

The figure below illustrates the modular structure of an MRE-based pricing framework.

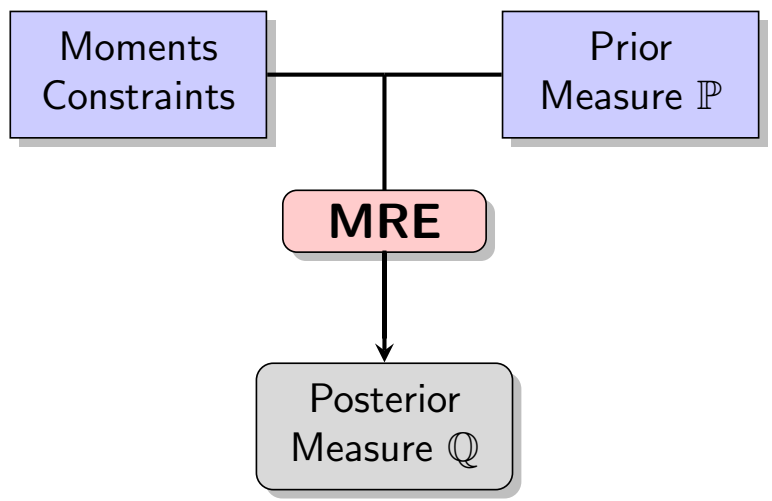


Figure 7: MRE pricing framework modules.

Moments Constraints and Prior Measure $\mathbb{P}$

The prior measure reflects the initial model, which should belong to one of the following categories:

i. an econometric model, i.e., physical measure, reflecting the most relevant properties of the financial variables driving the securities (e.g., short rates, equity prices, ...); and

ii. a reference pricing model in which case it will be compared with another, possibly, more complex model.

This latter aproach is useful for model risk quantification as proposed by Cont (2006), Cont and Deguest (2013), and Arrieta (2022).

On the other hand, the moment constraint block gathers the properties that the posterior measure will have.

Broadly, these constraints can be classified into two categories: benchmark market prices; and dummy derivatives prices.

For example, most liquid instruments or securities for ensuring martingale property belong to the first class.



Posterior Probability Measure $\mathbb{Q}$

Posterior measure $\mathbb{Q}$ verifies by construction that $\mathbb{E}^{\mathbb{Q}}[g_j] = c_j$, for all $j = 1, \ldots, m$, i.e., all benchmarks are properly priced.

Another result that relates pricing and hedging is stated next.

Proposition 6 (MRE Target Derivative Decomposition)

Let $\mathbb{Q}$ be a measure solving MRE optimization problem (11)*, and let $h \neq g_j$ be a derivative, then under $\mathbb{Q}$*

$$h = \beta_0 + \sum_{j=1}^{m} \beta_j g_j + \varepsilon, \tag{14}$$

where β_j is least-squares linear regression coefficient of h regarding g_j, and β_0 is the expectation of the error term ε of such linear regression.

This is a consequence of the MRE methodology in which, under $\mathbb{Q}$, the sensitivity of $\mathbb{E}^{\mathbb{Q}}[h]$ with respect to the benchmark input price c_j is the least-squares linear regression coefficient of h regarding g_j.

The proof of the proposition is detailed in Avellaneda et al. (2001).



MRE Pricing Framework Properties

i. Proposition 6 allows decomposing hedgeable and non-hedgeable risks, as stated in equation (14).

ii. Exact fitting[1] calibration avoids arbitrages or redundant securities, e.g., it does not converge if any of the benchmark securities is a linear combination of the rest of securities.

iii. New moment constraints can be any derivative securities, exchange traded (ETD) or Over the Counter (OTC), or even dummy ones for hedging or stressing modeling assumptions.

iv. New modeling assumptions can be added sequentially so its marginal impact can be calculated.

v. Market probabilities can be compared with historical/physical if an econometric models is chosen as prior measure.

vi. It is possible to quantify model uncertainty by using the relative entropy value between compared measures.

[1] All constraints are strictly satisfied, Avellaneda et al. (2001) proposed an alternative calibration method relaxing this requirement.



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Stochastic Local Volatility Model

The SLV model is a hybrid model that combines features of both local volatility models and stochastic volatility models.

The local volatility component depends on both the current asset price and time, capturing the current smile/skew observed in market implied volatilities.

The stochastic volatility component reflects that the volatility itself follows a stochastic process modeled using a stochastic differential equation (e.g., Heston model).

Provides a more accurate fit to market data, i.e., current option prices, and can be used to price exotic options that depend on the path of the underlying asset price due to its ability to capture the forward smile dynamics.

Complex and computationally demanding, it comprises the efficient computation of both a deterministic local volatility $\sigma_{\mathrm{LV}}(K, t)$ and a conditional expectation $\mathbb{E}^{\mathbb{Q}}\left[\xi^2(v(t)) \middle| S(t) = K\right]$.



Minimum Relative Entropy Framework

MRE pricing framework is a method used to determine a pricing measure that is closest to a prior probability measure (e.g., real-world or a simpler pricing measure), subject to certain constraints.

The *closeness* is measured using the concept of relative entropy (also known as Kullback-Leibler divergence).

The constraints, e.g., matching observed market prices, provide a systematic way to incorporate market information into the pricing model.

Proposition 6 allows a decomposition between hedgeable and non-hedgeable risk, thus it can handle incomplete markets.

The framework is modular, Figure 7, and the independence between the input blocks, ensures great robustness.

Computationally intensive due to the need to solve an optimization problem. Requires accurate estimation of the real-world probability measure, which can be challenging.



Comparison of MRE and SLV Models (i)

	MRE	**SLV**
Foundations	minimum relative entropy constrained optimization	hybrid local and stochastic volatility model
Goal	closest pricing measure to the real-world measure subject to matching market prices	matching observed market prices and accurately capture the forward smile dynamics
Complexity	optimization problem which can be computationally intensive, needs an estimation of the real-world measure	calibration of the local volatility function and the stochastic volatility parameters to market data and unobserved parameters
Market Fit	perfect if exact fitting approach is used	perfect if conditional expectation component is properly estimated



Comparison of MRE and SLV Models (ii)

	MRE	**SLV**
Advantages	splits hedgeable and non-hedgeable risks, handles incomplete markets, and provides a systematic way to incorporate market information	accurate fit to market data and sensible forward smile dynamics, no need to take into account real-world measure, i.e., keeps market complete
Limitations	needs an accurate estimation of the real-world measure, perfect market fit is not guaranteed if exact fitting approach is not used, computationally intensive in some cases	local vol assumes infinity strikes, unobserved stochastic volatility process parameters estimation, hedging in volatility is feasible, challenging conditional expectation estimation



References I

 Arrieta, Daniel (July 2015). "Minimum Relative Entropy and Cliquet Hedging". In: *Wilmott* 78. DOI: 10.1002/wilm.10435.

 — (Aug. 2022). "Model risk quantification based on relative entropy". In: *Journal of Risk Model Validation* 16(3), pp. 1–19. DOI: 10.21314/JRMV.2022.020.

 Avellaneda, Marco (1998). "Minimum Relative Entropy Calibration of Asset Pricing Models". In: *International Journal of Theoretical and Applied Finance* 01.04.

 Avellaneda, Marco et al. (2001). "Weighted Monte Carlo: A new technique for calibrating asset-pricing models". In: *International Journal of Theoretical and Applied Finance* 04.01.

References II

 Barndorff-Nielsen, Ole E. (1997). "Processes of normal inverse Gaussian type". In: *Finance and Stochastics* 2.1, pp. 41–68.

 Bates, David S. (1996). "Jumps and Stochastic Volatility: Exchange Rate Processes Implicit in Deutsche Mark Options". In: *The Review of Financial Studies* 9.1, pp. 69–107. DOI: 10.1093/rfs/9.1.69.

 Baxter and Rennie (2012). *Financial calculus. An introduction to derivative pricing.* Cambridge University Press.

 Black, Fisher and Myron Scholes (1973). "The Pricing of Options and Corporate Liabilities". In: *Journal of Political Economy* 81, pp. 637–654.

References III

 Breeden, Douglas T. and Robert H. Litzenberger (1978). "Prices of State-Contingent Claims Implicit in Option Prices". In: *Journal of Business* 51.4, pp. 621–651.

 Capinski, Marek and Ekkehard Kopp (2012). *Discrete Models of Financial Markets*. Ed. by Cambridge. Mastering Mathematical Finance. Cambridge University Press.

 Capinski, Marek, Ekkehard Kopp, and Janusz Traple (2012). *Stochastic Calculus for Finance*. Ed. by Cambridge. Mastering Mathematical Finance. Cambridge University Press.

 Cont, Rama (2006). "Model uncertainty and its impact on the pricing of derivative instruments". In: *Mathematical Finance* 16.3, pp. 519–547. DOI: 10.1111/j.1467-9965.2006.00281.x.

References IV


Cont, Rama and Romain Deguest (2013). "Equity correlations implied by index options: estimation and model uncertainty analysis". In: *Mathematical Finance* 23.3, pp. 496–530. DOI: 10.1111/j.1467-9965.2011.00503.x.


Cover, Thomas M. and Joy A. Thomas (2006). *Elements of information theory*. Second edition. John Wiley and Sons, Incorporated.


Cox, John C., Jonathan E. Ingersoll, and Stephen A. Ross (1985). "A theory of the term structure of interest rates". In: *Econometrica* 53.2, pp. 385–407.


Delbaen, Freddy and Walter Schachermayer (Sept. 1994). "A general version of the fundamental theorem of asset pricing". In: *Mathematische Annalen* 300, pp. 463–520. DOI: 10.1007/BF01450498.

References V

 Dupire, Bruno (1994). "Pricing with a smile". In: *Risk* 7, pp. 18–20.

 Elices, Alberto and Eduard Gimenez (May 2006). "Weighted Monte Carlo". In: *Risk Magazine* 19.5, pp. 78–83.

 Elliott, Robert J. and P. Ekkehard Kopp (2004). *Mathematics of Financial Markets*. Ed. by Springer. Springer Finance.

 Feller, W. (1951). "Two singular diffusion problems". In: *The Annals of Mathematics* 54, pp. 173–182.

 Gatheral, Jim (2004). "A parsimonious arbitrage-free implied volatility parameterization with application to the valuation of volatility derivatives". In: *Presentation at Global Derivatives and Risk Management*. Madrid.

References VI

 Gatheral, Jim, Thibault Jaisson, and Mathieu Rosenbaum (2018). "Volatility is rough". In: *Quantitative Finance* 18.6, pp. 933–949.

 Hagan, Patrick S. et al. (July 2002). "Managing Smile Risk". In: *Wilmott Magazine*, pp. 84–108.

 Harrison, J.M. and D.M. Kreps (1979). "Martingales and arbitrage in multi-period securities markets.". In: *Journal of Economic Theory* 20, pp. 381–408.

 Harrison, J.M. and S.R. Pliska (Aug. 1981). "Martingales and stochastic integrals in the theory of continuous trading". In: *Stochastic Processes and their Applications* 11.3, pp. 215–260.

References VII

 Harrison, J.M. and S.R. Pliska (Aug. 1983). "A stochastic calculus model of continuous trading: complete markets.". In: *Stochastic Processes and their Applications* 15.3, pp. 313–316.

 Heston, Steven L. (1993). "A closed-form solution for options with stochastic volatility with applications to bond and currency options". In: *The Review of Financial Studies* 6.2, pp. 327–343.

 Hull, John and Alan White (1987). "The pricing of options on assets with stochastic volatility". In: *The Journal of Finance* 42.2, pp. 281–300.

 Jex, Mark, Richard Henderson, and Dan Wang (1999). "Pricing Exotics Under the Smile". In: *Risk Magazine* 12.11, pp. 72–75.

References VIII

 Karatzas, Ioannis and Steven E. Shreve (1991). *Brownian motion and stochastic calculus*. Springer New York.

 Lipton, Alexander (2002). "The Vol Smile Problem". In: *Risk Magazine* 15.2, pp. 61–66.

 Lipton, Alexander and William McGhee (2002). "Universal Barriers". In: *Risk Magazine* 15.5, pp. 81–85.

 Madan, Dilip B., Peter P. Carr, and Eric C. Chang (Apr. 1998). "The Variance Gamma Process and Option Pricing". In: *Review of Finance* 2.1, pp. 79–105.

 Merton, Robert C. (1976). "Option pricing when underlying stock returns are discontinuous". In: *Journal of Financial Economics* 3.1-2, pp. 125–144.

References IX


Oosterlee, Cornelis W. and Lech A. Grzelak (2020). *Mathematical Modeling and Computation in Finance: With Exercises and Python and MATLAB Computer Codes*. World Scientific. ISBN: 978-1-78634-794-7.


Schöbel, R. and J. Zhu (1999). "Stochastic Volatility with an Ornstein-Uhlenbeck Process: An Extension". In: *European Finance Review* 3, pp. 23–46.


Stein, Elias M. and Jeremy C. Stein (1991). "Stock Price Distributions with Stochastic Volatility". In: *Review of Financial Studies* 4.4, pp. 727–752.

References X

 Stoep, A. W. van der, L. A. Grzelak, and C. W. Oosterlee (2014). "The Heston Stochastic-Local Volatility Model: Efficient Monte Carlo Simulation". In: *International Journal of Theoretical and Applied Finance* 17.7, p. 1450045. DOI: 10.1142/S0219024914500458.

 Taqqu, Murad S. and Walter Willinger (1987). "The Analysis of Finite Security Markets Using Martingales". In: *Advances in Applied Probability* 19.1, pp. 1–25. ISSN: 00018678. DOI: 10.2307/1427371.